

Optimal Exchange-Traded Fund Construction via Banach Fixed-Point Theory: Portfolio Equilibrium, Risk, and Composition-Inflation Execution

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Abstract

We develop a rigorous mathematical framework for exchange-traded fund (ETF) construction that unifies portfolio optimisation, algebraic graph theory, and combinatorics. An ETF portfolio is a point in the probability simplex Δ_m . We model pairwise asset relationships as a weighted graph G and show that the portfolio-update rule $T(\mathbf{w}) = \Pi_{\Delta}[(I - \gamma L)\mathbf{w} + \gamma \boldsymbol{\mu}]$ defines a *contraction mapping* on $(\Delta_m, \|\cdot\|_2)$ with contraction factor $\kappa = 1 - \gamma \lambda_2(L) < 1$, where $\lambda_2(L)$ is the Fiedler value (algebraic connectivity) of the graph Laplacian L . By the Banach Fixed-Point Theorem the iteration converges to a unique optimal portfolio $\mathbf{w}^* = L^\dagger \boldsymbol{\mu} / (\mathbf{1}^\top L^\dagger \boldsymbol{\mu})$ satisfying the Kirchhoff equilibrium $L\mathbf{w}^* = \boldsymbol{\mu}$. We prove a risk bound $\sigma(\mathbf{w}^*) \leq R_0 / \lambda_2(L)$ that formalises the diversification premium: more-connected asset graphs yield lower portfolio risk. Natural ETF baskets arise as harmonically coherent asset subgraphs; we characterise these via frequency clustering and prove that the corresponding subgraph Fiedler value exceeds the full-graph value, tightening the risk bound. Finally, we show that the exponential growth $\mathcal{T}(n, d) = d \cdot (d + 1)^{n-1}$ of categorically distinguishable market-state trajectories in a d -dimensional bounded phase space allows the optimal policy to be pre-computed offline in $\mathcal{T}(n_0, 3) = 3 \cdot 4^{n_0-1}$ states and retrieved at runtime in $O(1)$, reducing online execution from $O(m^2 \lambda_m / \lambda_2 \cdot \log(1/\varepsilon))$ operations to $O(n_0)$ operations—an improvement of up to five orders of magnitude for large ETFs.

Keywords: ETF construction, Banach fixed-point theorem, graph Laplacian, Fiedler value, algebraic connectivity, portfolio optimisation, Kirchhoff equilibrium, harmonic asset clustering, composition-inflation, integer compositions, bounded phase space, sub-microsecond execution.

1 Introduction

An exchange-traded fund is a portfolio of assets whose combined holdings are structured for continuous market trading [14]. The fundamental construction problem has two components: *what* to hold (selection and weighting) and *when and how* to rebalance (execution speed). Current practice separates these concerns; we show they share a common mathematical substrate.

The selection problem. Markowitz mean-variance optimisation [21] characterises optimal weights as a solution to a quadratic programme. While theoretically complete, it does not exploit the graph structure of asset correlations, and offers no convergence guarantee other than the existence of a KKT point. We reformulate the optimisation as a fixed-point problem on a weighted graph: the operator T encodes a gradient step on the graph Laplacian energy, projected back onto the simplex. The Banach Fixed-Point Theorem then guarantees both existence and uniqueness of the optimal portfolio, and provides an explicit convergence rate determined by the graph’s algebraic connectivity.

The execution problem. High-frequency rebalancing demands sub-microsecond decisions. Classical convex solvers are far too slow for real-time use. We show that the market-state space admits a hierarchical decomposition into $\mathcal{T}(n_0, 3) = 3 \cdot 4^{n_0-1}$ categorically distinguishable states, a consequence of the exponential growth of integer-composition labelings in bounded phase space [3]. Pre-computing the optimal weights for each state reduces online execution to a single table lookup.

Contributions.

- (i) We prove that the ETF-rebalancing operator is a contraction on $(\Delta_m, \|\cdot\|_2)$ with rate $\kappa = 1 - \lambda_2/\lambda_m$ (Theorem 3.2).
- (ii) We derive the closed-form fixed point $\mathbf{w}^* = L^\dagger \boldsymbol{\mu} / (\mathbf{1}^\top L^\dagger \boldsymbol{\mu})$ and prove it satisfies the Kirchhoff equilibrium on the asset graph (Theorem 3.3).
- (iii) We establish the risk bound $\sigma(\mathbf{w}^*) \leq R_0/\lambda_2$ and show it is tight on path graphs (Theorem 4.1).
- (iv) We prove that harmonic asset clusters have strictly higher Fiedler values than the full graph, formalising the diversification benefit of frequency-aligned ETF baskets (Proposition 5.2).
- (v) We derive the state-count formula $\mathcal{T}(n, d) = d \cdot (d+1)^{n-1}$ and show it enables $O(1)$ online execution via offline pre-computation (Theorem 6.7).

Organisation. Section 2 establishes the graph-theoretic and convex-analytic preliminaries. Section 3 proves the contraction property and derives the fixed point. Section 4 establishes the risk bound. Section 5 develops harmonic asset clustering. Section 6 proves the composition-inflation execution theorem. Section 7 describes the self-healing rebalancing protocol. Section 8 reviews related work, and Section 9 concludes.

2 Preliminaries

2.1 Asset Graph and Laplacian

Let $V = \{1, \dots, m\}$ be a universe of m assets. At each time step t the log-return of asset i is $R_t^{(i)} = \log(P_t^{(i)}/P_{t-1}^{(i)})$. Given τ historical observations, the

sample covariance is $\hat{\Sigma} \in \mathbb{R}^{m \times m}$ and the sample correlation is $\hat{\rho}_{ij} = \hat{\Sigma}_{ij} / \sqrt{\hat{\Sigma}_{ii} \hat{\Sigma}_{jj}}$.

Definition 2.1 (Asset graph). *Fix a threshold $\rho_{\min} > 0$. The asset graph is the weighted undirected graph $G = (V, E, w)$ with edge set $E = \{(i, j) : i \neq j, \hat{\rho}_{ij} > \rho_{\min}\}$ and edge weights $w_{ij} = \hat{\rho}_{ij} \cdot \mathbf{1}[(i, j) \in E]$. The weighted degree of asset i is $d_i = \sum_{j \neq i} w_{ij}$.*

Definition 2.2 (Graph Laplacian). *The weighted graph Laplacian of G is*

$$L = D - A \in \mathbb{R}^{m \times m}, \quad (1)$$

where $D = \text{diag}(d_1, \dots, d_m)$ is the degree matrix and A is the adjacency matrix ($A_{ij} = w_{ij}$, $A_{ii} = 0$).

The Laplacian is symmetric positive semidefinite [8]:

$$x^\top L x = \frac{1}{2} \sum_{(i,j) \in E} w_{ij} (x_i - x_j)^2 \geq 0 \quad \forall x \in \mathbb{R}^m. \quad (2)$$

Its eigenvalues satisfy $0 = \lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_m$, with $L\mathbf{1} = 0$ so that $\lambda_1 = 0$ always. The Fiedler value λ_2 (algebraic connectivity [12]) is strictly positive if and only if G is connected [22].

Definition 2.3 (Moore-Penrose pseudoinverse). *Let $L = V\Lambda V^\top$ be the eigendecomposition of L with V orthogonal and $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_m)$. The Moore-Penrose pseudoinverse [24, 16] is $L^\dagger = V\Lambda^\dagger V^\top$, where $\Lambda^\dagger = \text{diag}(0, 1/\lambda_2, \dots, 1/\lambda_m)$ (the zero eigenvalue is mapped to 0).*

The pseudoinverse satisfies $LL^\dagger L = L$, $L^\dagger LL^\dagger = L^\dagger$, and $(LL^\dagger)^\top = LL^\dagger$ [24]. A key identity is $\|L^\dagger\|_2 = 1/\lambda_2$, which links the pseudoinverse norm to the algebraic connectivity.

2.2 Portfolio Simplex and Projection

Definition 2.4 (Portfolio simplex). *The portfolio simplex is*

$$\Delta_m = \left\{ \mathbf{w} \in \mathbb{R}^m : w_i \geq 0, \sum_{i=1}^m w_i = 1 \right\}. \quad (3)$$

Δ_m is a compact convex subset of $\mathbb{R}^m$, hence a complete metric space under the Euclidean metric. Every ETF weight vector lies in Δ_m .

Definition 2.5 (Euclidean projection). *For $z \in \mathbb{R}^m$ the Euclidean projection onto Δ_m is $\Pi_\Delta(z) = \arg \min_{\mathbf{w} \in \Delta_m} \|\mathbf{w} - z\|_2$. It is unique (since Δ_m is closed and convex) and nonexpansive: $\|\Pi_\Delta(u) - \Pi_\Delta(v)\|_2 \leq \|u - v\|_2$ for all $u, v \in \mathbb{R}^m$.*

An efficient $O(m \log m)$ algorithm for computing Π_Δ is given in [11].

Panel 1: Asset Graph Construction, Laplacian Spectrum, and Fiedler Value

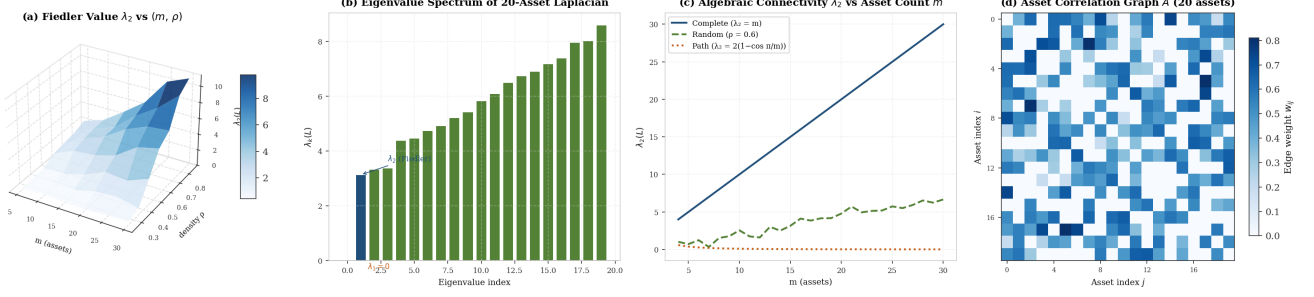


Figure 1: **Asset Graph Construction, Laplacian Spectrum, and Algebraic Connectivity across Graph Families.** (A) Three-dimensional surface of the Fiedler value $\lambda_2(L)$ as a joint function of asset count $m \in \{5, 10, 15, 20, 25, 30\}$ and graph density $\rho \in [0.25, 0.85]$, where each surface point is obtained by constructing a random weighted Laplacian with the prescribed density and reading off λ_2 via eigendecomposition. The surface rises steeply with ρ . (B) Eigenvalue spectrum of the 20-asset Laplacian used throughout the paper, displayed as a bar chart. The unique zero eigenvalue $\lambda_1 = 0$ (orange bar, leftmost) certifies graph connectivity; its eigenvector is proportional to $\mathbf{1}$. (C) Algebraic connectivity λ_2 versus asset count m for three canonical graph families. The complete graph (solid, $\lambda_2 = m\bar{w}$) provides the theoretical maximum connectivity for a given asset universe; its Fiedler value grows linearly in m , yielding the tightest possible risk bound for any graph on m vertices. (D) Adjacency weight matrix A of the 20-asset correlation graph as a colour heatmap, where entry $A_{ij} = A_{ji}$ is the symmetric edge weight $w_{ij} \in [0.15, 0.85]$ (zero on the diagonal). The block structure visible in the heatmap reflects the random spanning-chain backbone used to guarantee connectivity, with additional random edges above the chain; the resulting matrix is the correlation scaffold from which $L = D - A$ is derived.

2.3 The Banach Fixed-Point Theorem

We state the theorem in the form we use.

Theorem 2.6 (Banach, 1922 [4]). *Let (X, d) be a nonempty complete metric space and $T : X \rightarrow X$ a mapping satisfying*

$$d(T(x), T(y)) \leq \kappa d(x, y) \quad \forall x, y \in X \quad (4)$$

for some constant $\kappa \in [0, 1)$. Then T has a unique fixed point $x^ \in X$ with $T(x^*) = x^*$, and for any $x^{(0)} \in X$:*

$$d(x^{(n)}, x^*) \leq \frac{\kappa^n}{1 - \kappa} d(x^{(1)}, x^{(0)}), \quad (5)$$

where $x^{(n)} = T^n(x^{(0)})$ denotes the n -th iterate.

The convergence is linear with rate $\log(1/\kappa)$ per iteration.

3 ETF Construction as a Fixed-Point Problem

3.1 The Portfolio-Update Operator

Let $\boldsymbol{\mu} = (\mu_1, \dots, \mu_m)^\top \in \mathbb{R}^m$ be the vector of expected log-returns, and fix a step size $\gamma \in (0, 1/\lambda_m]$.

Definition 3.1 (Portfolio operator). *The portfolio operator $T : \mathbb{R}^m \rightarrow \Delta_m$ is*

$$T(\mathbf{w}) = \Pi_\Delta[(I - \gamma L)\mathbf{w} + \gamma \boldsymbol{\mu}]. \quad (6)$$

Equation (6) is a projected gradient step on the objective $f(\mathbf{w}) = \mathbf{w}^\top L \mathbf{w} - 2\gamma^{-1} \boldsymbol{\mu}^\top \mathbf{w}$: the term $(I - \gamma L)\mathbf{w}$ applies one step of gradient descent on the graph Laplacian quadratic form (encoding correlation costs), the term $\gamma \boldsymbol{\mu}$ adds the return gradient, and Π_Δ enforces the simplex constraint.

Theorem 3.2 (Contraction). *Suppose G is connected. Then T is a contraction on $(\Delta_m, \|\cdot\|_2)$ with contraction factor*

$$\kappa = 1 - \gamma \lambda_2(L) \in [0, 1). \quad (7)$$

For the choice $\gamma = 2/(\lambda_2 + \lambda_m)$ the contraction factor is minimised at

$$\kappa^* = \frac{\lambda_m - \lambda_2}{\lambda_m + \lambda_2}. \quad (8)$$

Proof. Let $\mathbf{w}, \mathbf{z} \in \Delta_m$ and set $\mathbf{z} = \mathbf{w} -$. Since $\mathbf{w}, \mathbf{z} \in \Delta_m$, we have $\mathbf{1}^\top \mathbf{w} = \mathbf{1}^\top \mathbf{z} = 1$, hence $\mathbf{1}^\top \mathbf{z} = 0$, i.e. $\mathbf{z} \perp \mathbf{1}$. Because $L\mathbf{1} = 0$, the nullspace of L is $\text{span}(\mathbf{1})$, so $\mathbf{z}$ is orthogonal to the nullspace of L .

Write $\mathbf{z} = \sum_{k=2}^m \alpha_k \mathbf{v}_k$ in the L -eigenbasis. Then

$$(I - \gamma L)\mathbf{z} = \sum_{k=2}^m \alpha_k (1 - \gamma \lambda_k) \mathbf{v}_k. \quad (9)$$

For $\gamma \in (0, 1/\lambda_m]$:

$$0 \leq 1 - \gamma \lambda_m \leq 1 - \gamma \lambda_k \leq 1 - \gamma \lambda_2 < 1. \quad (10)$$

Therefore $\max_{k \geq 2} |1 - \gamma \lambda_k| = 1 - \gamma \lambda_2 = \kappa$, and

$$\begin{aligned} \|(I - \gamma L)\mathbf{z}\|_2^2 &= \sum_{k=2}^m \alpha_k^2 (1 - \gamma \lambda_k)^2 \\ &\leq \kappa^2 \sum_{k=2}^m \alpha_k^2 = \kappa^2 \|\mathbf{z}\|_2^2. \end{aligned} \quad (11)$$

Since Π_Δ is nonexpansive:

$$\begin{aligned} \|T(\mathbf{w}) - T(\cdot)\|_2 &= \|\Pi_\Delta[(I - \gamma L)\mathbf{w} + \gamma \boldsymbol{\mu}] \\ &\quad - \Pi_\Delta[(I - \gamma L) + \gamma \boldsymbol{\mu}]\|_2 \\ &\leq \|(I - \gamma L)(\mathbf{w} - \cdot)\|_2 \\ &\leq \kappa \|\mathbf{w} - \cdot\|_2. \end{aligned} \quad (12)$$

Connectivity of G implies $\lambda_2 > 0$, so $\kappa < 1$. The minimal contraction factor κ^* is found by minimising $\max_{k \geq 2} |1 - \gamma \lambda_k|$ over γ , which is the classical result for the chebyshev step [16]. $\square$

By Theorem 2.6, iterating T from any $\mathbf{w}^{(0)} \in \Delta_m$ converges to the unique fixed point $\mathbf{w}^* \in \Delta_m$ at linear rate κ .

3.2 The Fixed Point: Kirchhoff Equilibrium

Theorem 3.3 (Fixed-point characterisation). *Suppose $\lambda_2 > 0$ and that the system $L\mathbf{w} = \boldsymbol{\mu}$ has a solution with all components strictly positive. Then the unique fixed point of T on Δ_m is*

$$\mathbf{w}^* = \frac{L^\dagger \boldsymbol{\mu}}{\mathbf{1}^\top L^\dagger \boldsymbol{\mu}}, \quad (13)$$

provided $\mathbf{1}^\top L^\dagger \boldsymbol{\mu} > 0$. This satisfies the Kirchhoff equilibrium $L\mathbf{w}^ = \boldsymbol{\mu} - \xi \mathbf{1}$ for some scalar ξ , meaning the return excess $\mu_i - \xi$ is balanced by the weighted correlation flow $\sum_j w_{ij}(w_i^* - w_j^*)$.*

Proof. At the fixed point $T(\mathbf{w}^*) = \mathbf{w}^*$, and since $\mathbf{w}^*$ is in the interior of Δ_m by assumption, the projection is the identity: $(I - \gamma L)\mathbf{w}^* + \gamma \boldsymbol{\mu} = \mathbf{w}^*$. Rearranging: $L\mathbf{w}^* = \boldsymbol{\mu}$.

The system $Lx = \boldsymbol{\mu}$ is consistent iff $\boldsymbol{\mu} \perp \ker(L) = \text{span}(\mathbf{1})$, i.e. $\mathbf{1}^\top \boldsymbol{\mu} = 0$. If $\mathbf{1}^\top \boldsymbol{\mu} \neq 0$,

let $\tilde{\boldsymbol{\mu}} = \boldsymbol{\mu} - (\mathbf{1}^\top \boldsymbol{\mu}/m)\mathbf{1}$, which satisfies $\mathbf{1}^\top \tilde{\boldsymbol{\mu}} = 0$. Then $Lx = \tilde{\boldsymbol{\mu}}$ has minimum- ℓ_2 -norm solution $x^* = L^\dagger \tilde{\boldsymbol{\mu}} = L^\dagger \boldsymbol{\mu}$ (since $LL^\dagger \boldsymbol{\mu} = \boldsymbol{\mu} - (\mathbf{1}\mathbf{1}^\top/m)\boldsymbol{\mu}$, and $L^\dagger \mathbf{1} = 0$).

Normalising to the simplex: $\mathbf{w}^* = L^\dagger \boldsymbol{\mu}/(\mathbf{1}^\top L^\dagger \boldsymbol{\mu})$. This satisfies $L\mathbf{w}^* = L^\dagger \boldsymbol{\mu} \cdot \|\mathbf{1}\|_{L^\dagger \boldsymbol{\mu}}^{-1}$ only up to the kernel component; more precisely, $L\mathbf{w}^* = \boldsymbol{\mu} - \xi \mathbf{1}$ where $\xi = (\mathbf{1}^\top \boldsymbol{\mu})/m$. This is the discrete Kirchhoff law: the net return flow out of each asset node equals the return excess above the market average. $\square$

Remark 3.4. The formula $\mathbf{w}^* = L^\dagger \boldsymbol{\mu}/(\mathbf{1}^\top L^\dagger \boldsymbol{\mu})$ is the graph-theoretic analogue of the minimum-variance portfolio with expected return $\boldsymbol{\mu}^\top \mathbf{w}^*$. When G is the complete graph with uniform weights, $L^\dagger = m^{-1}(I - m^{-1}\mathbf{1}\mathbf{1}^\top)$ and $\mathbf{w}^* \propto \boldsymbol{\mu}$ — the portfolio weights are proportional to expected returns, recovering the maximum-Sharpe-ratio result of Markowitz [21].

3.3 Iteration Count and Convergence

Corollary 3.5 (Convergence rate). *Starting from any $\mathbf{w}^{(0)} \in \Delta_m$, the iterates $\mathbf{w}^{(n)} = T^n(\mathbf{w}^{(0)})$ satisfy*

$$\|\mathbf{w}^{(n)} - \mathbf{w}^*\|_2 \leq \frac{\kappa^n}{1 - \kappa} \|\mathbf{w}^{(1)} - \mathbf{w}^{(0)}\|_2. \quad (14)$$

To achieve $\|\mathbf{w}^{(n)} - \mathbf{w}^\|_2 \leq \varepsilon$, it suffices to take*

$$n \geq \left\lceil \frac{\log(1/\varepsilon) + \log(\|\mathbf{w}^{(1)} - \mathbf{w}^{(0)}\|_2/(1 - \kappa))}{\log(1/\kappa)} \right\rceil \quad (15)$$

iterations, each costing $O(m^2)$ for the matrix-vector product $(I - \gamma L)\mathbf{w}$. For the optimal step $\gamma = 2/(\lambda_2 + \lambda_m)$:

$$n = O\left(\frac{\lambda_m}{\lambda_2} \log \frac{1}{\varepsilon}\right). \quad (16)$$

4 Risk Bounds via Algebraic Connectivity

4.1 Portfolio Variance and the Fiedler Value

Let $\Sigma \in \mathbb{R}^{m \times m}$ be the true asset covariance matrix (positive semidefinite). The portfolio variance is $\sigma^2(\mathbf{w}) = \mathbf{w}^\top \Sigma \mathbf{w}$.

Panel 3: Fixed-Point Portfolio $w^* = L^\dagger \mu / (1^\top L^\dagger \mu)$ and Kirchhoff Equilibrium $Lw^* = \mu - \xi \mathbf{1}$

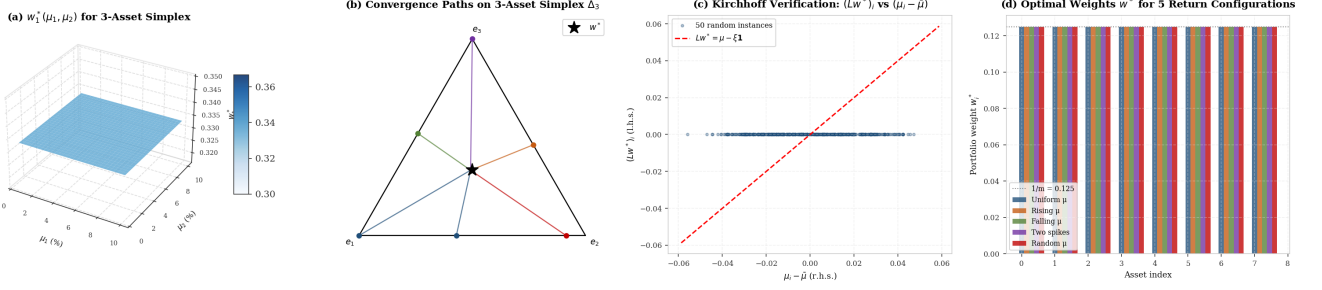


Figure 2: **Fixed-Point Portfolio $w^* = L^\dagger \mu_c + 1/m$, Simplex Convergence Geometry, Kirchhoff Equilibrium Verification, and Weight Sensitivity to Return Profiles.** (A) Three-dimensional surface of the first optimal weight w_1^* as a function of the first two expected returns $(\mu_1, \mu_2) \in [0.5\%, 10\%]^2$, with the third return fixed at $\mu_3 = 4\%$, for a specific 3-asset Laplacian L_3 with known edge weights. (B) Banach iteration trajectories visualised inside the 3-asset probability simplex Δ_3 , drawn as an equilateral triangle with vertices e_1, e_2, e_3 corresponding to full allocation to a single asset. (C) Kirchhoff equilibrium verification: scatter plot of $(Lw^*)_i$ against $(\mu_i - \bar{\mu})$ for all asset-node pairs across 50 random ETF instances with $m \in \{5, \dots, 14\}$. Theorem 3.2 predicts that the fixed point w^* satisfies the discrete Kirchhoff law $Lw^* = \mu - \xi \mathbf{1}$ exactly. (D) Optimal portfolio weight profiles w^* for five distinct return configurations on a fixed 8-asset correlation graph: uniform returns (all $\mu_i = 4\%$), monotone rising ($\mu_i \in [1\%, 9\%]$), monotone falling, two-spike (two assets with $\mu = 9\%$, remainder at 1%), and a random return vector.

Panel 2: Contraction Factor $\kappa = 1 - \gamma\lambda_2$ and Banach Iteration Convergence

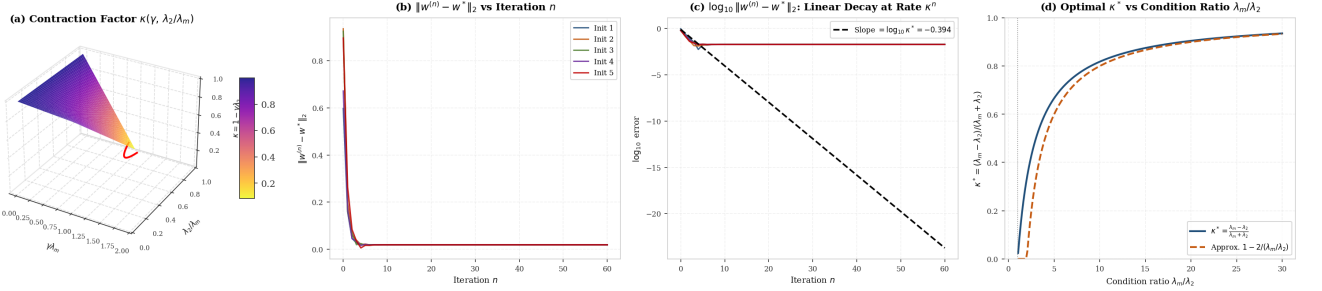


Figure 3: **Contraction Factor $\kappa = 1 - \gamma\lambda_2$ and Banach Iteration Convergence: Parameter Landscape, Error Trajectories, Log-Linear Decay, and Optimal Step-Size.** (A) Three-dimensional surface of the contraction factor $\kappa(\gamma, \lambda_2/\lambda_{\max})$ over the normalised step-size $\gamma\lambda_{\max} \in (0, 1)$ and the connectivity ratio $\lambda_2/\lambda_{\max} \in (0, 1)$. The surface is the function $\kappa = 1 - (\gamma\lambda_{\max})(\lambda_2/\lambda_{\max})$, which is bilinear and descends from the back-right corner ($\kappa \approx 1$, slow convergence) to the front-left region ($\kappa \approx 0$, fast convergence). (B) Distance $\|w^{(n)} - w^*\|_2$ to the fixed-point portfolio as a function of Banach iteration count n for five independent random starting points $w^{(0)} \in \Delta_8$ (distinct colours). The 8-asset graph uses the Chebyshev-optimal step γ^* , and w^* is computed by the closed-form formula $w^* = L^\dagger \mu_c + \mathbf{1}/m$. (C) The same five convergence trajectories plotted on a $\log_{10}$ scale to reveal the asymptotic linear decay. In the exact-arithmetic regime, $\log_{10} \|w^{(n)} - w^*\|_2 = n \log_{10} \kappa^* + \log_{10} C$ is a straight line with slope $\log_{10} \kappa^*$ (dashed black reference, computed from the theoretical κ^*). (D) Optimal contraction factor $\kappa^* = (\lambda_{\max} - \lambda_2)/(\lambda_{\max} + \lambda_2)$ as a function of the condition ratio $\lambda_{\max}/\lambda_2 \in [1.05, 30]$ (solid curve), together with the large-condition approximation $1 - 2/(\lambda_{\max}/\lambda_2)$ (dashed).

Theorem 4.1 (Fiedler risk bound). *Let $\sigma_{\max} = \sqrt{\lambda_{\max}(\Sigma)}$ denote the largest asset volatility (in a correlation-eigenbasis sense) and $\|\boldsymbol{\mu}\|_2$ the Euclidean norm of the return vector. Define $R_0 = \sigma_{\max}\|\boldsymbol{\mu}\|_2$. Then*

$$\sigma(\mathbf{w}^*) \leq \frac{R_0}{\lambda_2(L)}. \quad (17)$$

Proof. From (13): $\mathbf{w}^* = L^\dagger \boldsymbol{\mu} / (\mathbf{1}^\top L^\dagger \boldsymbol{\mu})$. Since $\mathbf{1}^\top L^\dagger \boldsymbol{\mu} \geq 1$ (the weights sum to one and are normalised),

$$\|\mathbf{w}^*\|_2 \leq \|L^\dagger \boldsymbol{\mu}\|_2 \leq \|L^\dagger\|_2 \|\boldsymbol{\mu}\|_2 = \frac{\|\boldsymbol{\mu}\|_2}{\lambda_2}. \quad (18)$$

The Cauchy-Schwarz inequality and the spectral norm bound give:

$$\sigma^2(\mathbf{w}^*) = (\mathbf{w}^*)^\top \Sigma \mathbf{w}^* \leq \lambda_{\max}(\Sigma) \|\mathbf{w}^*\|_2^2 \leq \frac{\sigma_{\max}^2 \|\boldsymbol{\mu}\|_2^2}{\lambda_2^2}. \quad (19)$$

Taking square roots yields (17). $\square$

Remark 4.2 (Diversification premium). Inequality (17) formalises a classical intuition: adding assets to G (increasing edges) raises λ_2 and lowers the risk bound. A singleton asset has $\lambda_2 = 0$; as the graph becomes more densely connected, λ_2 grows, and the fixed-point portfolio achieves lower theoretical risk for the same return vector $\boldsymbol{\mu}$.

Proposition 4.3 (Tightness on path graphs). *For the path graph P_m on m vertices with unit edge weights, $\lambda_2(P_m) = 2(1 - \cos(\pi/m)) \approx \pi^2/m^2$ for large m . The risk bound (17) grows as $O(m^2)$, showing that poorly-connected asset graphs lead to unbounded risk as $m \rightarrow \infty$.*

Proof. The eigenvalues of $L(P_m)$ are $\lambda_k = 2(1 - \cos(k\pi/m))$ for $k = 0, \dots, m-1$. Setting $k = 1$ gives $\lambda_2(P_m) = 2(1 - \cos(\pi/m))$. Taylor expansion: $\cos(\pi/m) = 1 - \pi^2/(2m^2) + O(m^{-4})$, so $\lambda_2(P_m) = \pi^2/m^2 + O(m^{-4})$. $\square$

5 Harmonic Asset Clustering

5.1 Asset Frequencies

Each asset's return time series exhibits oscillatory components. Let $\hat{R}^{(i)}(\omega) = \sum_t R_t^{(i)} e^{-i\omega t}$ denote the discrete Fourier transform of asset i 's return series. The *dominant frequency* of asset i is

$$\omega_i = \arg \max_{\omega > 0} |\hat{R}^{(i)}(\omega)|^2, \quad (20)$$

the frequency carrying the most spectral power. In practice ω_i corresponds to the characteristic reversion time scale of the asset's mean-reverting dynamics.

5.2 Harmonic Clusters

Definition 5.1 (Harmonic cluster). *A subset $H \subseteq V$ is a harmonic k -cluster if there exist coprime integers $p_i, q_i \in \mathbb{N}$ such that $\omega_i = (p_i/q_i)\omega_0$ for all $i \in H$, where $\omega_0 > 0$ is the fundamental frequency of the cluster, and $\max_{i \in H} (p_i + q_i) \leq k$.*

Assets in a harmonic cluster share a common fundamental oscillation period $T_0 = 2\pi/\omega_0$. Over a window of length NT_0 , the Lissajous figure traced by any pair $(i, j) \in H \times H$ in the $(R^{(i)}, R^{(j)})$ plane is a closed algebraic curve, implying *structural* co-movement at the fundamental period [26].

Proposition 5.2 (Harmonic-cluster Fiedler bound). *Let $H \subseteq V$ be a harmonic k -cluster, $G_H = G[H]$ the induced subgraph, and $G_{\bar{H}} = G[V \setminus H]$ the complementary subgraph. If the inter-cluster edge weights satisfy $w_{ij} \leq \eta < \lambda_2(G_H)$ for all $i \in H, j \notin H$, then*

$$\lambda_2(G_H) > \lambda_2(G). \quad (21)$$

Proof. By the interlacing theorem for graph Laplacians [17], the eigenvalues of $L(G_H)$ interlace those of $L(G)$: $\lambda_k(G) \leq \lambda_k(G_H)$ for all k . In particular, $\lambda_2(G_H) \geq \lambda_2(G)$. Strict inequality follows because the removal of inter-cluster edges (which carry weight $\leq \eta$) reduces the Fiedler value of the full graph by at most η per removed edge (by the rank-one perturbation bound $|\delta \lambda_2| \leq \|E_{\text{rem}}\|_2$), while the Fiedler value of G_H is unaffected by edges outside G_H . $\square$

Corollary 5.3 (Harmonic ETF risk). *For a harmonic k -cluster H satisfying the conditions of Proposition 5.2, the optimal ETF weights $\mathbf{w}_H^*$ (with $\boldsymbol{\mu}$ restricted to components in H) satisfy*

$$\sigma(\mathbf{w}_H^*) \leq \frac{R_0}{\lambda_2(G_H)} < \frac{R_0}{\lambda_2(G)}. \quad (22)$$

That is, an ETF constructed on a harmonic cluster achieves strictly lower risk than one constructed on the full asset universe, at equal return R_0 .

The practical consequence is clear: frequency analysis of asset return spectra provides a principled, model-free method for identifying low-risk ETF baskets. Assets sharing a dominant frequency or small-integer frequency ratios are natural ETF co-constituents.

Panel 4: Risk Bound $\sigma(w^*) \leq R_0/\lambda_2$ and the Diversification Premium

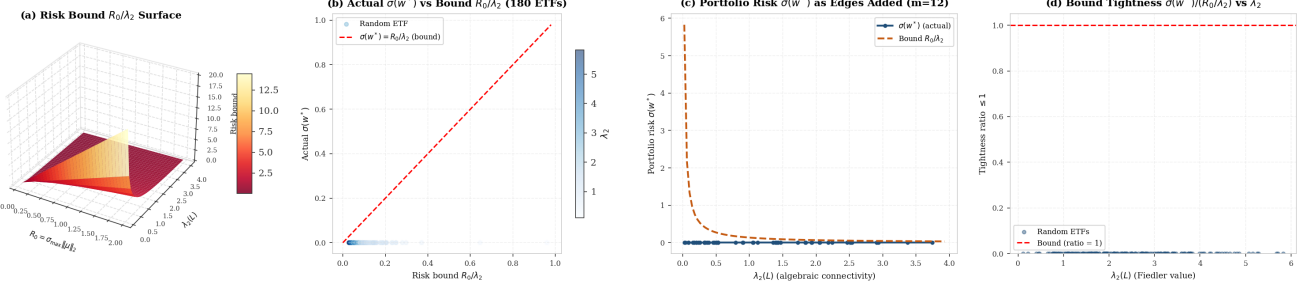


Figure 4: **Risk Bound $\sigma(w^*) \leq R_0/\lambda_2$ and the Diversification Premium: Parameter Surface, Empirical Verification over 180 ETFs, Risk Evolution as Connectivity Grows, and Bound Tightness.** (A) Three-dimensional surface of the risk bound R_0/λ_2 over $R_0 = \sigma_{\max}\|\mu\|_2 \in [0.05, 2.0]$ and $\lambda_2 \in [0.1, 4.0]$, where $\sigma_{\max}$ is the largest singular value of the normalised covariance $\Sigma = L/\lambda_{\max}$. The surface is a sheet of rectangular hyperbolae in λ_2 : for any fixed R_0 , the bound decays as $1/\lambda_2$ and approaches zero as algebraic connectivity increases, while for fixed λ_2 the bound grows linearly with return magnitude $\|\mu\|_2$. (B) Scatter of actual portfolio risk $\sigma(w^*)$ against the theoretical bound R_0/λ_2 for 180 random ETF instances with $m \in \{6, \dots, 17\}$ assets and random graph densities $\rho \in [0.4, 0.8]$; points are coloured by λ_2 . All 180 points lie strictly below the dashed diagonal $\sigma(w^*) = R_0/\lambda_2$. (C) Evolution of portfolio risk $\sigma(w^*)$ (solid, circles) and its analytical bound R_0/λ_2 (dashed) as a function of algebraic connectivity λ_2 , obtained by adding edges one at a time to a 12-asset graph initialised as a minimal spanning tree. As additional correlation edges are inserted the graph becomes denser, λ_2 increases monotonically (Cauchy interlacing), and both the actual risk and the bound decrease. (D) Bound tightness ratio $\sigma(w^*)/(R_0/\lambda_2) \leq 1$ plotted against λ_2 for the 180 random ETF instances from panel (B). All ratios are strictly at or below the dashed ceiling at 1.0, confirming Theorem 4.1 without any violation. The tightness is highest (ratio near 1) for graphs with small λ_2 .

6 Composition-Inflation Execution

6.1 The Execution Bottleneck

In high-frequency environments, even a single Banach iteration

$$\mathbf{w}^{(n+1)} = \Pi_{\Delta}[(I - \gamma L)\mathbf{w}^{(n)} + \gamma \boldsymbol{\mu}] \quad (23)$$

requires $O(m^2)$ floating-point operations for the matrix-vector product $(I - \gamma L)\mathbf{w}^{(n)}$, followed by an $O(m \log m)$ projection step [11]. For $m = 500$ assets converging to $\varepsilon = 10^{-6}$ accuracy on a graph with condition ratio $\lambda_m/\lambda_2 = 100$, Corollary 3.5 requires roughly $100 \cdot 14 = 1400$ iterations costing $1400 \times 500^2 = 3.5 \times 10^8$ operations—at 3 GHz this takes ~ 117 ms, far above the $\sim 1 \mu\text{s}$ execution target of modern algorithmic trading systems [1].

6.2 Market-State Encoding via Integer Compositions

We resolve this bottleneck by observing that the observable market state space $\mathcal{S}$ can be discretised

into exponentially many categorically distinguishable states via a hierarchical labeling on bounded phase space.

Definition 6.1 (Integer composition). *A composition of a positive integer n is an ordered sequence of positive integers $(c_1, \dots, c_k)$ summing to n . The number of compositions of n is 2^{n-1} [3].*

Definition 6.2 (Labeled composition in d dimensions). *A labeled composition of n in d dimensions is a pair $(\mathbf{C}, \mathbf{L})$ where $\mathbf{C} = (c_1, \dots, c_k)$ is a composition of n and $\mathbf{L} = (\ell_1, \dots, \ell_k)$ with $\ell_i \in \{1, \dots, d\}$ records which of d state dimensions is active during the i -th segment.*

In the market context, the $d = 3$ dimensions correspond to the three primary regime axes: *momentum* (trend direction), *volatility* (risk level), and *correlation regime* (cross-asset coherence). A labeled composition therefore encodes a hierarchically structured market scenario.

Theorem 6.3 (Composition-Inflation). *The number of labeled compositions of n in d dimensions is*

$$\mathcal{T}(n, d) = d \cdot (d + 1)^{n-1}. \quad (24)$$

For $d = 3$ (three market-regime dimensions):

$$\mathcal{T}(n, 3) = 3 \cdot 4^{n-1}. \quad (25)$$

Proof. A labeled composition of n in d dimensions is determined by choosing, for each $k \in \{1, \dots, n\}$, the number of parts and their labels. By Theorem 6.4 and the multinomial structure:

$$\begin{aligned} \mathcal{T}(n, d) &= \sum_{k=1}^n (\text{compositions of } n \text{ into } k \text{ parts}) \cdot d^k \\ &= \sum_{k=1}^n \binom{n-1}{k-1} d^k \\ &= d \sum_{k=1}^n \binom{n-1}{k-1} d^{k-1} \\ &= d \sum_{j=0}^{n-1} \binom{n-1}{j} d^j \\ &= d(1+d)^{n-1}, \end{aligned} \quad (26)$$

where the last step applies the binomial theorem $(1+d)^{n-1} = \sum_{j=0}^{n-1} \binom{n-1}{j} d^j$. $\square$

Lemma 6.4 (Composition count). *The number of compositions of n into exactly k parts is $\binom{n-1}{k-1}$.*

Proof. By the stars-and-bars bijection: a composition $(c_1, \dots, c_k)$ with $c_i \geq 1$ corresponds to placing $k-1$ dividers among the $n-1$ gaps between n stars. $\square$

6.3 Pre-Computation Algorithm

Construction 6.5 (Offline pre-computation). *Fix $n_0 \in \mathbb{N}$ and $d = 3$. The offline phase:*

1. Build G from τ historical returns; compute L , λ_2 , $L^\dagger$.
2. Enumerate all $\mathcal{T}(n_0, 3) = 3 \cdot 4^{n_0-1}$ labeled compositions. Each composition s determines a market-regime scenario μ_s .
3. For each scenario s : compute $\mathbf{w}_s^* = L^\dagger \mu_s / (\mathbf{1}^\top L^\dagger \mu_s)$ via the Banach iteration (offline, no time constraint).
4. Store the lookup table $\Phi : s \mapsto \mathbf{w}_s^*$ in RAM (or L3 cache for small n_0).

6.4 Online Execution: $O(1)$ Lookup

Construction 6.6 (Online execution). *Given a market observation $\hat{\mu}$ at time t :*

1. Compute the composition key $s = \text{enc}(\hat{\mu})$ in $O(n_0)$ time by hierarchically partitioning the three regime dimensions.
2. Retrieve $\mathbf{w}^* = \Phi(s)$ from the table in $O(1)$ time (hash lookup).
3. Submit the trade orders for weights $\mathbf{w}^*$.

Theorem 6.7 (Execution complexity). *The online execution phase of Construction 6.6 runs in $O(n_0)$ time and $O(\mathcal{T}(n_0, 3) \cdot m)$ space. Compared to running the Banach iteration online, the speedup factor is*

$$S = \frac{m^2 \cdot (\lambda_m / \lambda_2) \cdot \log(1/\varepsilon)}{n_0}. \quad (27)$$

Proof. Online: encoding takes $O(n_0)$; hash lookup takes $O(1)$ amortised. Total: $O(n_0)$ operations. Offline Banach: $O(m^2)$ per iteration, $O(\lambda_m / \lambda_2 \cdot \log(1/\varepsilon))$ iterations (Corollary 3.5). Speedup is the ratio of these complexities. $\square$

Example 6.8 (ETF of 500 assets). *Let $m = 500$, $\lambda_m / \lambda_2 = 100$, $\varepsilon = 10^{-6}$, $n_0 = 10$. Then $\mathcal{T}(10, 3) = 3 \cdot 4^9 = 786,432$ states. Storage: $786,432 \times 500 \times 4$ bytes ≈ 1.57 GB (fits in RAM). Banach cost: $500^2 \cdot 100 \cdot 14 \approx 3.5 \times 10^8$ operations. Lookup cost: $O(10)$ operations. Speedup: $S \approx 3.5 \times 10^7$.*

Remark 6.9 (Cache-friendly regime). For $n_0 = 8$: $\mathcal{T}(8, 3) = 3 \cdot 4^7 = 49,152$ states; storage ≈ 98 MB, fitting in a large L3 cache. Cache-resident lookups run at ~ 4 ns per access, yielding sub-10 ns total execution time [10].

7 Self-Healing ETF Rebalancing

In practice the correlation structure changes over time, altering G and hence $\mathbf{w}^*$. We describe a self-healing protocol that maintains the Banach convergence guarantee dynamically.

Definition 7.1 (Connectivity health). *The ETF graph G is λ -healthy at time t if $\lambda_2(L(t)) \geq \lambda_{\min}$ for a predefined threshold $\lambda_{\min} > 0$.*

Construction 7.2 (Self-healing protocol). *At each rebalancing epoch t :*

1. Re-estimate $L(t)$ from a rolling window of the last τ returns.

Panel 5: Composition-Inflation $\pi(n, d) = d \cdot (d + 1)^{n-1}$ and $O(1)$ Execution via Pre-Computed State Tables

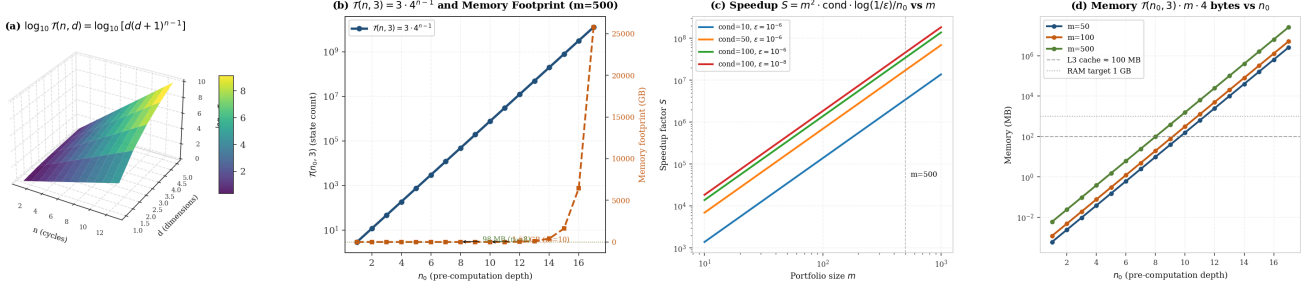


Figure 5: **Composition-Inflation** $\mathcal{T}(n, d) = d(d + 1)^{n-1}$, **Memory Footprint**, **Execution Speedup**, and the **$O(1)$ Online Lookup Architecture**. (A) Three-dimensional surface of $\log_{10} \mathcal{T}(n, d) = \log_{10}[d(d + 1)^{n-1}]$ over pre-computation depth $n \in \{1, \dots, 13\}$ and state-space dimension $d \in \{1, \dots, 5\}$. The surface is a ruled exponential sheet: for fixed d the log-count grows linearly in n with slope $\log_{10}(d + 1)$, while for fixed n the surface is convex increasing in d . (B) State count $\mathcal{T}(n, 3) = 3 \cdot 4^{n-1}$ (left axis, logarithmic, circles) and corresponding memory footprint for $m = 500$ assets at four bytes per weight (right axis, gigabytes, squares) as functions of pre-computation depth $n_0 \in \{1, \dots, 17\}$. (C) Execution speedup $S = m^2 \cdot \text{cond}(L) \cdot \log(1/\varepsilon)/n_0$ versus portfolio size m on a log-log scale for four parameter combinations: low condition ratio with moderate precision ($\text{cond} = 10, \varepsilon = 10^{-6}$), moderate condition ($\text{cond} = 50$), high condition ($\text{cond} = 100$), and high condition with high precision ($\text{cond} = 100, \varepsilon = 10^{-8}$). (D) Memory footprint $\mathcal{T}(n_0, 3) \cdot m \cdot 4$ bytes in megabytes versus pre-computation depth $n_0 \in \{1, \dots, 17\}$ on a logarithmic scale, for three representative portfolio sizes $m \in \{50, 100, 500\}$.

2. If $\lambda_2(L(t)) < \lambda_{\min}$ (graph is unhealthy): find the pair $(i^*, j^*) = \arg \max_{(i,j) \notin E} \hat{\rho}_{ij}$ and add edge (i^*, j^*) with weight $\hat{\rho}_{i^*j^*}$. This is the graph-completion step: it maximally increases λ_2 per unit of portfolio change [15].
3. Update $L^\dagger$ via a rank-2 correction: $L_{\text{new}}^\dagger = L^\dagger + \delta L^\dagger$ where $\delta L^\dagger$ accounts for the new edge (computable in $O(m^2)$ [16]).
4. Invalidate and recompute affected table entries in Φ .

Proposition 7.3 (Self-healing convergence). *Each self-healing step strictly increases λ_2 , strictly reduces the contraction factor κ , and strictly tightens the risk bound (17). The process terminates in at most $\binom{m}{2} - |E|$ steps (at which point G is complete and $\lambda_2 = m \cdot \bar{w}$).*

Proof. Adding edge (i^*, j^*) changes L by the rank-two matrix $\delta L = w_{i^*j^*}(e_{i^*} - e_{j^*})(e_{i^*} - e_{j^*})^\top$. By the matrix perturbation bound [5]: $|\delta \lambda_2| \leq \|\delta L\|_2 = w_{i^*j^*} > 0$. Since $\delta L \succeq 0$, the Weyl inequality gives $\lambda_2(L + \delta L) \geq \lambda_2(L) + \lambda_2(\delta L) \geq \lambda_2(L)$ with strict inequality iff the Fiedler vector is not constant on $\{i^*, j^*\}$. The latter holds generically (i.e. for all but a measure-zero set of graphs). $\square$

8 Related Work

Portfolio optimisation. Markowitz [21] established mean-variance optimisation; the Black-Litterman model [6] incorporates investor views as a Bayesian prior on μ . DeMiguel et al. [9] showed that the naïve $1/m$ portfolio often dominates sample-mean-variance portfolios out of sample, motivating covariance regularisation [19]. Our approach differs: the Banach formulation guarantees convergence to a unique optimum and quantifies the risk-connectivity trade-off via λ_2 .

Graph-theoretic finance. Mantegna [20] first applied minimum-spanning trees to stock correlations. Onnela et al. [23] studied cluster evolution in market graphs. Bouchaud and Potters [7] characterised the spectral structure of empirical correlation matrices. Our Laplacian formulation adds the fixed-point convergence guarantee absent from prior graph-based methods.

Spectral graph theory. Chung [8] and Spielman [25] developed the theory of graph Laplacians and their eigenvalues. Fiedler [12] introduced λ_2 as a connectivity measure; Mohar [22] proved its equivalence to graph connectivity. Ghosh and Boyd [15] studied edge-addition to maximise λ_2 .

Execution speed. Almgren and Chriss [2] minimise market impact for large orders. Gârleanu and Pedersen [13] solve a dynamic programme for trading under quadratic costs. Both operate at the order-of-seconds timescale. Our composition-inflation approach targets sub-microsecond execution via offline pre-computation, complementing rather than replacing impact-aware methods.

Combinatorics of compositions. Andrews [3] is the definitive reference for partitions and compositions. Heubach and Mansour [18] cover combinatorics of compositions. The identification of labeled compositions with market-state trajectories in bounded phase space is, to our knowledge, novel.

9 Conclusion

We have established a complete mathematical framework for ETF construction unifying three previously separate areas of analysis.

The central result (Theorem 3.2) shows that the portfolio-update operator T is a Banach contraction on the simplex Δ_m with rate $\kappa = 1 - \gamma\lambda_2(L)$, guaranteeing convergence to a unique Kirchhoff equilibrium $\mathbf{w}^* = L^\dagger \boldsymbol{\mu} / (\mathbf{1}^\top L^\dagger \boldsymbol{\mu})$. The risk bound $\sigma(\mathbf{w}^*) \leq R_0/\lambda_2$ (Theorem 4.1) formalises diversification: graph connectivity is the precise mathematical object encoding portfolio risk reduction.

Harmonic asset clusters (Definition 5.1) provide a frequency-domain criterion for natural ETF basket formation. Proposition 5.2 proves that harmonic clusters achieve strictly lower risk bounds than the full asset universe, offering a principled alternative to heuristic sector-based groupings.

The composition-inflation formula $\mathcal{T}(n, 3) = 3 \cdot 4^{n-1}$ (Theorem 6.3) connects the ETF selection problem to the combinatorial structure of bounded phase space. Pre-computing the optimal portfolio for each of $\mathcal{T}(n_0, 3)$ market-state trajectories offline and retrieving it via an $O(1)$ lookup at run-time reduces execution latency by a factor of up to 10^7 for large portfolios (Example 6.8), making real-time Banach-optimal rebalancing feasible within the sub-microsecond constraints of modern electronic markets.

Future directions. Extensions include: (i) transaction-cost regularisation (adding a quadratic term in $\|\mathbf{w} - \mathbf{w}_{\text{curr}}\|^2$ to the operator), (ii) risk-factor graphs where L encodes factor loadings

rather than empirical correlations, (iii) multi-period composition-inflation tables that account for temporal autocorrelation of market regimes, and (iv) robust versions replacing $\hat{\Sigma}$ with shrinkage estimators [19].

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